

IRE 317

Control System Engineering

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STABILITY ANALYSIS IN S-DOMAIN

What is Stability?

A system is said to be stable, if its output is under control. Otherwise, it is said to be unstable. A stable system produces a bounded (i.e., does not go to infinity or oscillate uncontrollably) output for a given bounded input.

Mathematically, this is called Bounded Input – Bounded Output (BIBO) Stability.

1. If you give a system a limited (finite) input then it should produce a limited (finite) output.
2. If the output grows without limit even for a small input, then the system **is unstable**.

Example

Stable: A spring–mass–damper system comes to rest after a disturbance.

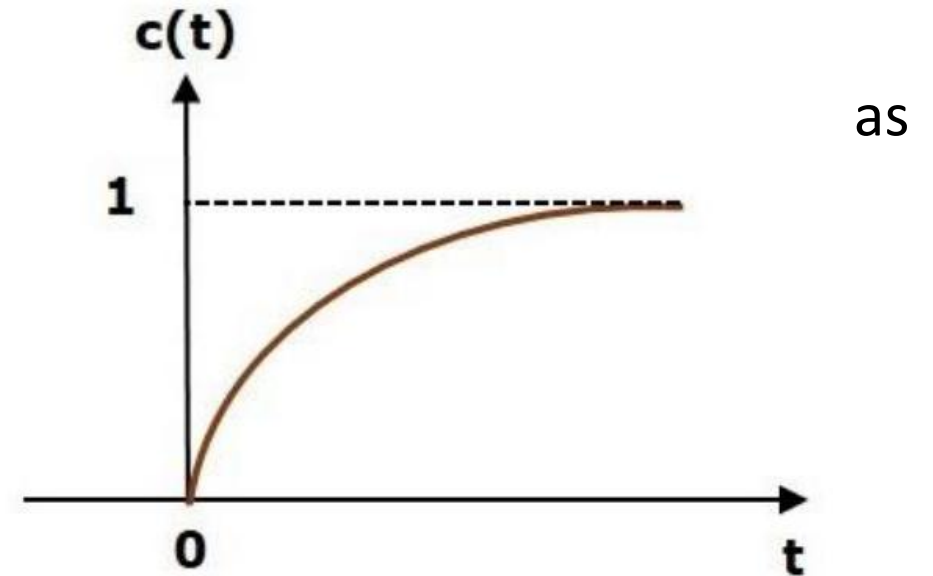
Unstable: A ball balanced on the top of a hill — a small disturbance causes it to roll away indefinitely.

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The following figure shows the response of a stable system. This is the response of first order control system for unit step input. This response has the values between 0 and 1. So, it is bounded output. We know that the unit step signal has the value of one for all positive values of t including zero. So, it is bounded input. Therefore, the first order control system is stable since both the input and the output are bounded.

Types of Systems based on Stability We can classify as follows.

1. Absolutely stable system
2. Conditionally stable system
3. Marginally stable system



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Absolutely Stable System

If the system is stable for all the range of system component values, then it is known as the absolutely stable system. The open loop control system is absolutely stable if all the poles of the open loop transfer function present in left half of 's' plane. Similarly, the closed loop control system is absolutely stable if all the poles of the closed loop transfer function present in the left half of the 's' plane.

Conditionally Stable System

If the system is stable for a certain range of system component values, then it is known as conditionally stable system.

Marginally Stable System

If the system is stable by producing an output signal with constant amplitude and constant frequency of oscillations for bounded input, then it is known as marginally stable system. The open loop control system is marginally stable if any two poles of the open loop transfer function is present on the imaginary axis. Similarly, the closed loop control system is marginally stable if any two poles of the closed loop transfer function is present on the imaginary axis.

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Type	Explanation	Example
Absolutely stable	Always under control	Fan at normal speeds, volume knob
Conditionally stable	Stable only in a safe range	Car at high speeds, mic-speaker distance
Marginally stable	Oscillates forever without growing or dying	Ideal pendulum, spinning top

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Routh-Hurwitz Stability Criterion

Routh-Hurwitz stability criterion is having one necessary condition and one sufficient condition for stability. If any control system doesn't satisfy the necessary condition, then we can say that the control system is unstable. But, if the control system satisfies the necessary condition, then it may or may not be stable. So, the sufficient condition is helpful for knowing whether the control system is stable or not.

Necessary Condition for Routh-Hurwitz Stability

The necessary condition is that the coefficients of the characteristic polynomial should be positive. This implies that all the roots of the characteristic equation should have negative real parts.

Consider the characteristic equation of the order 'n' is -

$$a_0s^n + a_1s^{n-1} + a_2s^{n-2} + \dots + a_{n-1}s^1 + a_ns^0 = 0$$

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Necessary Condition for Routh-Hurwitz Stability

The necessary condition is that the coefficients of the characteristic polynomial should be positive. This implies that all the roots of the characteristic equation should have negative real parts.

Take this characteristic equation:

$$s^2 + 5s + 6 = 0$$

All coefficients are positive:

$$1, 5, 6 > 0$$

Now find the roots:

$$s = \frac{-5 \pm \sqrt{25 - 24}}{2}$$
$$s = \frac{-5 \pm 1}{2}$$

So the roots are:

$$\bullet s_1 = -2$$

$$\bullet s_2 = -3$$

Both have negative real parts so it means a stable system

so, positive coefficients is the necessary condition to be satisfied

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Sufficient Condition for Routh-Hurwitz Stability

The sufficient condition is that all the elements of the first column of the Routh array should have the same sign. This means that all the elements of the first column of the Routh array should be either positive or negative.

Routh Array Method

If all the roots of the characteristic equation exist to the left half of the 's' plane, then the control system is stable. If at least one root of the characteristic equation exists to the right half of the 's' plane, then the control system is unstable. So, we have to find the roots of the characteristic equation to know whether the control system is stable or unstable. But, it is difficult to find the roots of the characteristic equation as order increases. So, to overcome this problem there we have the Routh array method. In this method, there is no need to calculate the roots of the characteristic equation. First formulate the Routh table and find the number of the sign changes in the first column of the Routh table. The number of sign changes in the first column of the Routh table gives the number of roots of characteristic equation that exist in the right half of the 's' plane and the control system is unstable.

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Follow this procedure for forming the Routh table.

1. Fill the first two rows of the Routh array with the coefficients of the characteristic polynomial as mentioned in the table below. Start with the coefficient of s_n and continue up to the coefficient of s_0 .
2. Fill the remaining rows of the Routh array with the elements as mentioned in the table below. Continue this process till you get the first column element of row s_0 is a_n . Here, a_n is the coefficient of s_0 in the characteristic polynomial.

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The following table shows the Routh array of the nth order characteristic polynomial.

$$a_0s^n + a_1s^{n-1} + a_2s^{n-2} + \dots + a_{n-1}s^1 + a_ns^0$$

s^n	a_0	a_2	a_4	a_6	...	...
s^{n-1}	a_1	a_3	a_5	a_7	...	...
s^{n-2}	b_1 $= \frac{a_1a_2 - a_3a_0}{a_1}$	b_2 $= \frac{a_1a_4 - a_5a_0}{a_1}$	b_3 $= \frac{a_1a_6 - a_7a_0}{a_1}$	...	...	...
s^{n-3}	c_1 $= \frac{b_1a_3 - b_2a_1}{b_1}$	c_2 $= \frac{b_1a_5 - b_3a_1}{b_1}$	$\vdots$			
$\vdots$	$\vdots$	$\vdots$	$\vdots$			
s^1	$\vdots$	$\vdots$				
s^0	a_n					

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Let us find the stability of the control system having characteristic equation,

$$s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$$

Step 1 – Verify the necessary condition for the Routh-Hurwitz stability.

All the coefficients of the characteristic polynomial,
 $s^4 + 3s^3 + 3s^2 + 2s + 1$ are positive. So, the control system satisfies the necessary condition.

Step 2 – Form the Routh array for the given characteristic polynomial.

s^4	1	3	1
s^3	3	2	
s^2	$\frac{(3 \times 3) - (2 \times 1)}{3} = \frac{7}{3}$	$\frac{(3 \times 1) - (0 \times 1)}{3} = \frac{3}{3} = 1$	
s^1	$\frac{\left(\frac{7}{3} \times 2\right) - (1 \times 3)}{\frac{7}{3}} = \frac{5}{7}$		
s^0	1		

Step 3 – Verify the sufficient condition for the Routh-Hurwitz stability. All the elements of the first column of the Routh array are positive. There is no sign change in the first column of the Routh array. So, the control system is stable.

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Special Cases of Routh Array

We may come across two types of situations, while forming the Routh table. It is difficult to

complete the Routh table from these two situations.

The two special cases are –

1. The first element of any row of the Routh's array is zero.
2. All the elements of any row of the Routh's array are zero.

Let us now discuss how to overcome the difficulty in these two cases, one by one. First Element of any row of the Routh's array is zero. If any row of the Routh's array contains only the first element as zero and at least one of the remaining elements have non-zero value, then replace the first element with a small positive integer, ϵ . And then continue the process of completing the Routh's table. Now, find the number of sign changes in the first column of the Routh's table by substituting ϵ tends to zero.

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Example

Let us find the stability of the control system having characteristic equation,

$$s^4 + 2s^3 + s^2 + 2s + 1 = 0$$

Step 1 – Verify the necessary condition for the Routh-Hurwitz stability.

All the coefficients of the characteristic polynomial,

$$s^4 + 2s^3 + s^2 + 2s + 1 = 0$$

are positive. So, the control system satisfied the necessary condition.

Step 2 – Form the Routh array for the given characteristic polynomial.

s^4	1	1	1
s^3	≥ 1	≥ 1	
s^2	$\frac{(1 \times 1) - (1 \times 1)}{1} = 0$	$\frac{(1 \times 1) - (0 \times 1)}{1} = 1$	
s^1			
s^0			

The row s^3 elements have 2 as the common factor. So, all these elements are divided by 2. Special case (i) – Only the first element of row s^2 is zero. So, replace it by ϵ and continue the process of completing the Routh table.

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s^4	1	1	1
s^3	1	1	
s^2	ϵ	1	
s^1	$\frac{(\epsilon \times 1) - (1 \times 1)}{\epsilon} = \frac{\epsilon - 1}{\epsilon}$		
s^0	1		

There are two sign changes in the first column of Routh table. Hence, the control system is unstable

s^4	1	1	1
s^3	1	1	
s^2	0	1	
s^1	$-\infty$		
s^0	1		

Step 3 – Verify the sufficient condition for the Routh-Hurwitz stability. As ϵ tends to zero, the Routh table becomes like this.

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All the Elements of any row of the Routh's array are zero

In this case, follow these two steps –

1. Write the auxiliary equation, $A(s)$ of the row, which is just above the row of zeros.
2. Differentiate the auxiliary equation, $A(s)$ with respect to s . Fill the row of zeros with these coefficients.

Example

Let us find the stability of the control system having characteristic equation,

$$s^5 + 3s^4 + s^3 + 3s^2 + s + 3 = 0$$

Step 1 – Verify the necessary condition for the Routh-Hurwitz stability.

All the coefficients of the given characteristic polynomial are positive. So, the control system satisfied the necessary condition.

Step 2 – Form the Routh array for the given characteristic polynomial.

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s^5	1	1	1
s^4	$\exists 1$	$\exists 1$	$\exists 1$
s^3	$\frac{(1 \times 1) - (1 \times 1)}{1} = 0$	$\frac{(1 \times 1) - (1 \times 1)}{1} = 0$	
s^2			
s^1			
s^0			

$$s^5 + 3s^4 + s^3 + 3s^2 + s + 3 = 0$$

The row s^4 elements have the common factor of 3. So, all these elements are divided by 3.

Special case (ii) – All the elements of row s^3 are zero. So, write the auxiliary equation, $A(s)$ of the row s^4 .

$$A(s) = s^4 + s^2 + 1$$

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$$\frac{dA(s)}{ds} = 4s^3 + 2s$$

Place these coefficients in row s^3 .

s^5	1	1	1
s^4	1	1	1
s^3	4 2	≥ 1	
s^2	$\frac{(2 \times 1) - (1 \times 1)}{2} = 0.5$	$\frac{(2 \times 1) - (0 \times 1)}{2} = 1$	
s^1	$\frac{(0.5 \times 1) - (1 \times 2)}{0.5} = \frac{-1.5}{0.5}$ $= -3$		
s^0	1		

Step 3 – Verify the sufficient condition for the Routh-Hurwitz stability.

There are two sign changes in the first column of Routh table. Hence, the control system is unstable.

In the Routh-Hurwitz stability criterion, we can know whether the closed loop poles are in on left half of the ‘s’ plane or on the right half of the ‘s’ plane or on an imaginary axis. So, we can’t find the nature of the control system. To overcome this limitation, there is a technique known as the root locus.